

Locality and the Vanishing Induction at Five and Six Points

We separate two statements that must be proved together.

- L_n : the full nonlocal top ansatz $B_n^{(n-3)}$, subject to the chosen $n-1$ row zeros, contains no crossing pair of poles.
- $V_n(k)$: an auxiliary projected coefficient $C_n^{(k)}$ is zero if it has

$$\nu(0) = 1, \quad \nu(k) = k + 2 \quad (k \geq 1) \quad (1)$$

independent usable row zeros.

The notation $C_n^{(k)}$ has one meaning throughout: it is an n -point function with exactly k simple pole factors in each term. A decorated symbol such as $C_{n;P,Q_1,\dots,Q_r}^{(k)}$ is the same class of object; the extra subscripts only record the parent P and the crossing poles used to isolate it. No additional letter distinguishes a compatible top piece from a constant residue piece. The letter B is used only for the original top ansatz. We use R_P only as shorthand for the operation $\text{Res}_{P=0}$ applied to the current function; it is not another ansatz class. A projected $C_n^{(k)}$ need not be a canonical lower-point amplitude, especially after a long-chord cut. The induction class includes all projected cut spaces with the same simple-pole grading and projected row zeros. We assume that the cyclic placement of a fixed number of independent usable zeros is immaterial. Accordingly, whenever a coefficient has r usable zeros, we represent them by the consecutive block $Z_1, \dots, Z_r$. This is the representative worst-case placement used throughout the auxiliary lemmas.

Throughout we assume square-free simple poles, constant numerators, and homogeneous degree equal to minus the number of pole slots. Consequently a pole-free remainder of negative degree is zero. We prove locality only: every crossing-free top coefficient $C_{n-1;P}^{(n-4)}$ is left arbitrary.

The residue mechanism and why strict zeros are needed

The induction works because a residue isolates a definite denominator sector. If F has a simple parent pole P , write

$$F = \frac{1}{P} \text{Res}_{P=0} F + F_{\text{reg } P}. \quad (2)$$

Thus proving $\text{Res}_{P=0} F = 0$ removes *every* term of F containing the parent pole P . More generally, an iterated residue

$$C_{m;P,Q_1,\dots,Q_s}^{(k)} := \text{Res}_{Q_s=0} \cdots \text{Res}_{Q_1=0} \text{Res}_{P=0} F \quad (3)$$

isolates the sector containing the displayed parent and crossing poles. If this C -function vanishes, that entire denominator sector is absent; one is not merely evaluating a single term at a special kinematic point.

The essential question is whether a zero of F is inherited by the residue. Suppose $F|_{Z_a} = 0$, and let $M = \{P, Q_1, \dots, Q_s\}$ be the pole support used in (3). If imposing Z_a together with $M = 0$ forces no further pole variable to vanish, then Z_a is *strict* on this support. Restriction to the row is then transverse to every unselected pole divisor, so restriction and the selected iterated residue commute:

$$\left(\text{Res}_M F \right) \Big|_{Z_a^{(M)}} = \text{Res}_M (F|_{Z_a}) = 0. \quad (4)$$

Here $Z_a^{(M)}$ denotes the row equations projected to the variables that remain after the residue. Equation (4) is the precise sense in which the lower $C_m^{(k)}$ *inherits* the zero. Only distinct projected row loci count as independent inherited zeros.

Strictness is necessary. If the same equations also force an unselected pole $Q = 0$, then the restricted locus lies on the extra Q -pole divisor. Terms with a Q pole can now participate in the vanishing of the full function, so the selected residue is not isolated. In that case (4) cannot be asserted: the ambient zero of F need not be a zero of $\text{Res}_M F$. In particular, killing only the deeper iterated coefficient $\text{Res}_Q \text{Res}_M F$ does not repair the problem, because other terms in the complete Q -parent sector can still pollute the row. The row becomes usable only after the complete statement

$$\text{Res}_{Q=0} F = 0 \quad (5)$$

has removed every term with a Q pole (and likewise for every additional forced pole). We call such a formerly polluted row *saturated-active*; logically it is usable because the polluting sectors have already been removed. Before that removal, only strict zeros may be inherited.

For a parent residue the practical expansion has the triangular form

$$\text{Res}_{P=0} F = C_{m;P}^{(r-1)} + \sum_{Q \in \text{Cr}(P)} \frac{C_{m;P,Q}^{(r-2)}}{Q} + \sum_{Q < Q'} \frac{C_{m;P,Q,Q'}^{(r-3)}}{QQ'} + \dots \quad (6)$$

One first studies the deepest crossing coefficients. For each displayed C -function, one counts only the rows strict on its full displayed support. Those inherited zeros allow the lower-point vanishing lemma to set that C -function to zero. One then moves upward through the expansion. If a previously polluted row becomes usable, this must follow from the vanishing of a complete polluting parent residue, not merely from the disappearance of one subresidue. Repeating this parent by parent eliminates every crossing sector, while the crossing-free top piece $C_{m;P}^{(r-1)}$ is not fixed in a proof of locality alone. The strict-zero tables below record exactly the inheritance data needed for this argument.

1 Five-point example

1.1 The full nonlocal ansatz and its four zeros

The five-point planar pole alphabet is

$$\mathcal{P}_5 = \{X_{13}, X_{14}, X_{24}, X_{25}, X_{35}\}. \quad (7)$$

No compatibility is imposed. The complete two-pole ansatz is

$$B_5^{(2)} = \sum_{\substack{S \subset \mathcal{P}_5 \\ |S|=2}} \frac{b_S}{\prod_{P \in S} P}. \quad (8)$$

It contains all ten unordered pole pairs. The four imposed row zeros are

$$Z_1 : X_{13} + X_{24} - X_{14} = 0, \quad X_{14} + X_{25} - X_{24} = 0, \quad (9)$$

$$Z_2 : X_{24} + X_{35} - X_{25} = 0, \quad X_{25} + X_{13} - X_{35} = 0, \quad (10)$$

$$Z_3 : X_{35} + X_{14} - X_{13} = 0, \quad X_{13} + X_{24} - X_{14} = 0, \quad (11)$$

$$Z_4 : X_{14} + X_{25} - X_{24} = 0, \quad X_{24} + X_{35} - X_{25} = 0. \quad (12)$$

We assume $B_5^{(2)}|_{Z_a} = 0$ for $a = 1, \dots, 4$.

Definition 1 (Strict and saturated-active zeros). *For an explicit pole support M , define its closure on the row Z_a by*

$$\text{cl}_a(M) := \{Q \in \mathcal{P}_n \mid (Z_a = 0, P = 0 \text{ for every } P \in M) \implies Q = 0\}. \quad (13)$$

In words, $\text{cl}_a(M)$ is the set of all pole variables that are forced to zero after imposing the row equations Z_a and setting every pole in M to zero. The row Z_a is strict if

$$\text{cl}_a(M) = M, \quad (14)$$

meaning precisely that no additional pole is forced to vanish. If $\text{cl}_a(M) \setminus M$ is nonempty, the row is polluted by the corresponding pole sectors. Suppose, for example, that $Q \in \text{cl}_a(M) \setminus M$. The row does not become usable merely because the iterated coefficient on $M \cup \{Q\}$ vanishes. It becomes saturated-active only after the complete polluting parent residue

$$\text{Res}_{Q=0} F = 0 \quad (15)$$

has been established for the current function F ; this removes every term with a Q pole. If several extra poles are forced, the complete parent residue of each must first vanish. A usable family consists of distinct projected row loci; repeated ambient labels or coincident projections are counted only once.

For example, take $M = \{X_{14}, X_{25}\}$ at five points. On Z_2 , setting $X_{14} = X_{25} = 0$ leaves

$$X_{24} + X_{35} = 0, \quad X_{13} - X_{35} = 0.$$

No additional individual pole is forced to zero, so $\text{cl}_2(M) = M$: Z_2 is strict. On Z_1 , however, the same substitution gives

$$X_{13} + X_{24} = 0, \quad X_{24} = 0,$$

and hence also $X_{13} = 0$. Therefore

$$\text{cl}_1(M) = \{X_{14}, X_{25}, X_{24}, X_{13}\} \supsetneq M.$$

The row Z_1 is polluted by the deeper X_{24} and X_{13} sectors. To activate it one would have to remove the complete X_{24} and X_{13} parent residues. Vanishing only the iterated coefficient supported on $\{X_{14}, X_{25}, X_{24}\}$, for example, would not suffice.

Lemma 1 (One-zero constant lemma). *A constant iterated-residue coefficient with one strict, or already saturated-active, zero vanishes.*

Proof. On the usable zero locus no unremoved pole sector contaminates the selected iterated residue. The residue of the vanishing parent function is therefore the constant itself, so that constant is zero. $\square$

1.2 Every parent residue

For a parent pole P , write

$$R_P := \operatorname{Res}_{P=0} B_5^{(2)} = C_{4;P}^{(1)} + \sum_{Q \in \operatorname{Cr}(P)} \frac{C_{4;P,Q}^{(0)}}{Q}. \quad (16)$$

Here $C_{4;P}^{(1)}$ contains the poles compatible with P and is not fixed. The five crossing sets are

$$\begin{array}{c|c} P & \operatorname{Cr}(P) \\ \hline X_{13} & X_{24}, X_{25} \\ X_{24} & X_{13}, X_{35} \\ X_{35} & X_{14}, X_{24} \end{array} \quad \begin{array}{c|c} P & \operatorname{Cr}(P) \\ \hline X_{14} & X_{25}, X_{35} \\ X_{25} & X_{13}, X_{14} \end{array} \quad (17)$$

and the complete strict-zero table is

parent P	crossing coefficient	strict zeros among $Z_1, \dots, Z_4$
X_{13}	$C_{4;13,24}^{(0)}$	Z_2, Z_4
X_{13}	$C_{4;13,25}^{(0)}$	Z_1, Z_3, Z_4
X_{14}	$C_{4;14,25}^{(0)}$	Z_2, Z_3
X_{14}	$C_{4;14,35}^{(0)}$	Z_1, Z_2, Z_4
X_{24}	$C_{4;24,13}^{(0)}$	Z_2, Z_4
X_{24}	$C_{4;24,35}^{(0)}$	Z_1, Z_3
X_{25}	$C_{4;25,13}^{(0)}$	Z_1, Z_3, Z_4
X_{25}	$C_{4;25,14}^{(0)}$	Z_2, Z_3
X_{35}	$C_{4;35,14}^{(0)}$	Z_1, Z_2, Z_4
X_{35}	$C_{4;35,24}^{(0)}$	Z_1, Z_3

The reversed entries are the same double residues: $C_{4;P,Q}^{(0)} = C_{4;Q,P}^{(0)}$. Thus the ten displayed entries represent the five distinct crossing pairs. Every one has at least two strict zeros, although Lemma 1 needs only one. Hence all crossing coefficients vanish. No saturated-active zero is needed.

It is important that this does not remove a parent pole. It proves only

$$\operatorname{Res}_{Q=0} \operatorname{Res}_{P=0} B_5^{(2)} = 0 \quad (P \text{ crosses } Q); \quad (18)$$

the crossing-free $C_{4;P}^{(1)}$ remains. The surviving denominator pairs are the five pentagon triangulations:

$$B_{5,\text{local}}^{(2)} = \frac{a_1}{X_{13}X_{14}} + \frac{a_2}{X_{14}X_{24}} + \frac{a_3}{X_{24}X_{25}} + \frac{a_4}{X_{25}X_{35}} + \frac{a_5}{X_{35}X_{13}}. \quad (19)$$

The a_i are arbitrary. This proves L_5 .

1.3 The auxiliary five-point vanishing lemma

The statement $V_5(0)$ is exactly Lemma 1. We now prove $V_5(1)$ using the representative consecutive starting zeros Z_1, Z_2, Z_3 .

Take a pole residue of

$$C_5^{(1)} = \sum_{P \in \mathcal{P}_5} \frac{c_P}{P}. \quad (20)$$

Each c_P is a constant coefficient. Its strict rows are

parent	strict rows	polluted rows and forced poles
X_{13}	Z_3	$Z_1 \rightarrow X_{25}; Z_2 \rightarrow X_{24}$
X_{14}	Z_1, Z_2, Z_3	none
X_{24}	Z_1	$Z_2 \rightarrow X_{13}; Z_3 \rightarrow X_{35}$
X_{25}	Z_2, Z_3	$Z_1 \rightarrow X_{13}$
X_{35}	Z_1, Z_2	$Z_3 \rightarrow X_{24}$

Every parent has at least one strict row among the consecutive starting triple. Lemma 1 therefore kills every pole residue. Since $C_5^{(1)}$ has degree -1 , it has no pole-free remainder:

$$\boxed{V_5(0) : 1 \text{ zero}, \quad V_5(1) : 3 \text{ zeros.}} \quad (21)$$

By the assumed insensitivity to cyclic placement, the conclusion depends only on the number three, not on the displayed row labels. This is the lower input used at six points.

2 Six-point example

2.1 The full nonlocal ansatz and five imposed zeros

The pole alphabet is

$$\mathcal{P}_6 = \{X_{13}, X_{14}, X_{15}, X_{24}, X_{25}, X_{26}, X_{35}, X_{36}, X_{46}\}. \quad (22)$$

The full nonlocal cubic ansatz contains all $\binom{9}{3} = 84$ triples:

$$B_6^{(3)} = \sum_{\substack{S \subset \mathcal{P}_6 \\ |S|=3}} \frac{b_S}{\prod_{P \in S} P}. \quad (23)$$

The three poles in a term are not assumed compatible. We impose $B_6^{(3)}|_{Z_a} = 0$ for $a = 1, \dots, 5$, where

$$Z_1 : X_{13} + X_{24} - X_{14} = 0, \quad X_{14} + X_{25} - X_{15} - X_{24} = 0, \quad X_{15} + X_{26} - X_{25} = 0, \quad (24)$$

$$Z_2 : X_{24} + X_{35} - X_{25} = 0, \quad X_{25} + X_{36} - X_{26} - X_{35} = 0, \quad X_{26} + X_{13} - X_{36} = 0, \quad (25)$$

$$Z_3 : X_{35} + X_{46} - X_{36} = 0, \quad X_{36} + X_{14} - X_{13} - X_{46} = 0, \quad X_{13} + X_{24} - X_{14} = 0, \quad (26)$$

$$Z_4 : X_{46} + X_{15} - X_{14} = 0, \quad X_{14} + X_{25} - X_{15} - X_{24} = 0, \quad X_{24} + X_{35} - X_{25} = 0, \quad (27)$$

$$Z_5 : X_{15} + X_{26} - X_{25} = 0, \quad X_{25} + X_{36} - X_{26} - X_{35} = 0, \quad X_{35} + X_{46} - X_{36} = 0. \quad (28)$$

Process the six short chords first:

$$\mathcal{S}_6 = \{X_{13}, X_{24}, X_{35}, X_{46}, X_{15}, X_{26}\}. \quad (29)$$

Their crossing sets are

$$\begin{array}{c|c|c|c}
P & \text{Cr}(P) & P & \text{Cr}(P) \\
\hline
X_{13} & X_{24}, X_{25}, X_{26} & X_{24} & X_{13}, X_{35}, X_{36} \\
X_{35} & X_{14}, X_{24}, X_{46} & X_{46} & X_{15}, X_{25}, X_{35} \\
X_{15} & X_{26}, X_{36}, X_{46} & X_{26} & X_{13}, X_{14}, X_{15}
\end{array} \tag{30}$$

2.2 All six short-parent residues

If $\text{Cr}(P) = \{Q_1, Q_2, Q_3\}$, the complete crossing expansion is

$$R_P := \text{Res}_{P=0} B_6^{(3)} = C_{5;P}^{(2)} + \sum_{i=1}^3 \frac{C_{5;P,Q_i}^{(1)}}{Q_i} + \sum_{1 \leq i < j \leq 3} \frac{C_{5;P,Q_i,Q_j}^{(0)}}{Q_i Q_j}. \tag{31}$$

The $C^{(0)}$ coefficients are removed first, followed by the $C^{(1)}$ coefficients. The full strict-zero data are:

parent	coefficient	strict zeros	lower lemma needed
X_{13}	$C_{5;13,24,25}^{(0)}$	Z_5	$V_5(0)$: one
X_{13}	$C_{5;13,24,26}^{(0)}$	Z_4, Z_5	$V_5(0)$: one
X_{13}	$C_{5;13,25,26}^{(0)}$	Z_3, Z_4	$V_5(0)$: one
X_{13}	$C_{5;13,24}^{(1)}$	Z_2, Z_4, Z_5	$V_5(1)$: three
X_{13}	$C_{5;13,25}^{(1)}$	Z_3, Z_4, Z_5	$V_5(1)$: three
X_{13}	$C_{5;13,26}^{(1)}$	Z_1, Z_3, Z_4, Z_5	$V_5(1)$: three
X_{24}	$C_{5;24,13,35}^{(0)}$	Z_5	$V_5(0)$: one
X_{24}	$C_{5;24,13,36}^{(0)}$	Z_4, Z_5	$V_5(0)$: one
X_{24}	$C_{5;24,35,36}^{(0)}$	Z_1	$V_5(0)$: one
X_{24}	$C_{5;24,13}^{(1)}$	Z_2, Z_4, Z_5	$V_5(1)$: three
X_{24}	$C_{5;24,35}^{(1)}$	Z_1, Z_3, Z_5	$V_5(1)$: three
X_{24}	$C_{5;24,36}^{(1)}$	Z_1, Z_4, Z_5	$V_5(1)$: three
X_{35}	$C_{5;35,14,24}^{(0)}$	Z_5	$V_5(0)$: one
X_{35}	$C_{5;35,14,46}^{(0)}$	Z_1, Z_2	$V_5(0)$: one
X_{35}	$C_{5;35,24,46}^{(0)}$	Z_1	$V_5(0)$: one
X_{35}	$C_{5;35,14}^{(1)}$	Z_1, Z_2, Z_5	$V_5(1)$: three
X_{35}	$C_{5;35,24}^{(1)}$	Z_1, Z_3, Z_5	$V_5(1)$: three
X_{35}	$C_{5;35,46}^{(1)}$	Z_1, Z_2, Z_4	$V_5(1)$: three
X_{46}	$C_{5;46,15,25}^{(0)}$	Z_2, Z_3	$V_5(0)$: one
X_{46}	$C_{5;46,15,35}^{(0)}$	Z_1, Z_2	$V_5(0)$: one
X_{46}	$C_{5;46,25,35}^{(0)}$	Z_1	$V_5(0)$: one
X_{46}	$C_{5;46,15}^{(1)}$	Z_1, Z_2, Z_3, Z_5	$V_5(1)$: three
X_{46}	$C_{5;46,25}^{(1)}$	Z_1, Z_2, Z_3	$V_5(1)$: three
X_{46}	$C_{5;46,35}^{(1)}$	Z_1, Z_2, Z_4	$V_5(1)$: three
X_{15}	$C_{5;15,26,36}^{(0)}$	Z_3, Z_4	$V_5(0)$: one
X_{15}	$C_{5;15,26,46}^{(0)}$	Z_2, Z_3	$V_5(0)$: one
X_{15}	$C_{5;15,36,46}^{(0)}$	Z_1, Z_2	$V_5(0)$: one
X_{15}	$C_{5;15,26}^{(1)}$	Z_2, Z_3, Z_4	$V_5(1)$: three

parent	coefficient	strict zeros	lower lemma needed
X_{15}	$C_{5;15,36}^{(1)}$	Z_1, Z_2, Z_3, Z_4	$V_5(1)$: three
X_{15}	$C_{5;15,46}^{(1)}$	Z_1, Z_2, Z_3, Z_5	$V_5(1)$: three
X_{26}	$C_{5;26,13,14}^{(0)}$	Z_4, Z_5	$V_5(0)$: one
X_{26}	$C_{5;26,13,15}^{(0)}$	Z_3, Z_4	$V_5(0)$: one
X_{26}	$C_{5;26,14,15}^{(0)}$	Z_2, Z_3	$V_5(0)$: one
X_{26}	$C_{5;26,13}^{(1)}$	Z_1, Z_3, Z_4, Z_5	$V_5(1)$: three
X_{26}	$C_{5;26,14}^{(1)}$	Z_2, Z_3, Z_4, Z_5	$V_5(1)$: three
X_{26}	$C_{5;26,15}^{(1)}$	Z_2, Z_3, Z_4	$V_5(1)$: three

Thus every short-parent $C_5^{(0)}$ has at least one strict zero and every $C_5^{(1)}$ has at least three distinct strict projected zeros. Equations (21) kill all of them. In particular,

$$\text{Res}_{Q=0} \text{Res}_S B_6^{(3)} = 0 \quad (S \in \mathcal{S}_6, Q \in \text{Cr}(S)). \quad (32)$$

This removes every mixed sector containing a crossing pair with a short chord; it does not remove the short parent or $C_{5;S}^{(2)}$. All six short parents are marginal in the sharp counting because some $C_5^{(1)}$ reaches exactly three strict zeros. None is bad, and no activation is used.

2.3 The three long-parent residues

The remaining poles are the long triangle

$$\mathcal{L}_6 = \{X_{14}, X_{25}, X_{36}\}. \quad (33)$$

After (32), the short crossing chords can be omitted from each long-parent expansion. The three reduced residues are

$$R_{14} = C_{5;14}^{(2)} + \frac{C_{5;14,25}^{(1)}}{X_{25}} + \frac{C_{5;14,36}^{(1)}}{X_{36}} + \frac{C_{5;14,25,36}^{(0)}}{X_{25}X_{36}}, \quad (34)$$

$$R_{25} = C_{5;25}^{(2)} + \frac{C_{5;25,14}^{(1)}}{X_{14}} + \frac{C_{5;25,36}^{(1)}}{X_{36}} + \frac{C_{5;25,14,36}^{(0)}}{X_{14}X_{36}}, \quad (35)$$

$$R_{36} = C_{5;36}^{(2)} + \frac{C_{5;36,14}^{(1)}}{X_{14}} + \frac{C_{5;36,25}^{(1)}}{X_{25}} + \frac{C_{5;36,14,25}^{(0)}}{X_{14}X_{25}}. \quad (36)$$

Every one of these nine displayed coefficients has all five rows $Z_1, \dots, Z_5$ strict. The $C^{(0)}$ coefficients vanish by $V_5(0)$ and the $C^{(1)}$ coefficients by $V_5(1)$. Therefore the three remaining crossing pairs

$$\{X_{14}, X_{25}\}, \quad \{X_{14}, X_{36}\}, \quad \{X_{25}, X_{36}\} \quad (37)$$

also vanish. Combining the two shells gives

$$\text{Res}_{Q=0} \text{Res}_{P=0} B_6^{(3)} = 0 \quad (P \text{ crosses } Q). \quad (38)$$

Every surviving triple is a hexagon triangulation:

$$\boxed{B_{6,\text{local}}^{(3)} = \sum_{T \in \mathcal{T}_6} \frac{a_T}{\prod_{P \in T} P}, \quad |\mathcal{T}_6| = 14.} \quad (39)$$

Again the a_T are arbitrary. This proves $\mathbf{L}_6$ from the five-point vanishing statements.

2.4 The auxiliary five-to-six induction

We now prove the auxiliary statements needed after the six-point locality calculation. We use the consecutive representatives Z_1, Z_2, Z_3 for $C_6^{(1)}$ and Z_1, Z_2, Z_3, Z_4 for $C_6^{(2)}$.

If a piece has too few strict zeros, it is not killed immediately: one must first eliminate the complete parent residue responsible for a polluted row and then return to it. This produces the parent-to-parent propagation below. The grade-zero statement $V_6(0)$ is the universal one-zero constant lemma, Lemma 1.

Three zeros kill $C_6^{(1)}$

Start from Z_1, Z_2, Z_3 . For each parent pole P , the coefficient to be killed is $C_{5;P}^{(0)} = \text{Res}_{P=0} C_6^{(1)}$. The table names that coefficient explicitly. A polluted entry $Z_a \rightarrow Q$ means that, on the support $P = 0$, the row Z_a also forces the extra pole $Q = 0$.

parent P	constant residue of $C_6^{(1)}$	strict zeros of this constant	polluted rows at P
X_{13}	$C_{5;13}^{(0)} = \text{Res}_{X_{13}=0} C_6^{(1)}$	Z_3	$Z_1 \rightarrow X_{26}; Z_2 \rightarrow X_{24}$
X_{14}	$C_{5;14}^{(0)} = \text{Res}_{X_{14}=0} C_6^{(1)}$	Z_1, Z_2, Z_3	none
X_{15}	$C_{5;15}^{(0)} = \text{Res}_{X_{15}=0} C_6^{(1)}$	Z_1, Z_2, Z_3	none
X_{24}	$C_{5;24}^{(0)} = \text{Res}_{X_{24}=0} C_6^{(1)}$	Z_1	$Z_2 \rightarrow X_{13}; Z_3 \rightarrow X_{35}$
X_{25}	$C_{5;25}^{(0)} = \text{Res}_{X_{25}=0} C_6^{(1)}$	Z_1, Z_2, Z_3	none
X_{26}	$C_{5;26}^{(0)} = \text{Res}_{X_{26}=0} C_6^{(1)}$	Z_2, Z_3	$Z_1 \rightarrow X_{13}$
X_{35}	$C_{5;35}^{(0)} = \text{Res}_{X_{35}=0} C_6^{(1)}$	Z_1, Z_2	$Z_3 \rightarrow X_{24}$
X_{36}	$C_{5;36}^{(0)} = \text{Res}_{X_{36}=0} C_6^{(1)}$	Z_1, Z_2, Z_3	none
X_{46}	$C_{5;46}^{(0)} = \text{Res}_{X_{46}=0} C_6^{(1)}$	Z_1, Z_2, Z_3	none

Every named constant residue has at least one strict zero. Hence $V_5(0)$ kills all nine $C_{5;P}^{(0)}$ directly; the polluted rows are not needed. Since all pole residues of $C_6^{(1)}$ vanish and its degree is -1 , it has no pole-free remainder. Therefore

$$\boxed{V_6(1) : C_6^{(1)} = 0 \text{ from three consecutive zeros.}} \quad (40)$$

Four zeros kill $C_6^{(2)}$

This proof is the $C^{(2)}$ analogue of the $B_6^{(3)}$ locality proof, with one fewer pole slot. We again process the six short chords first and expand each parent residue only in chords crossing that parent.

Step 1: short parent and crossing expansion. Let $P \in \mathcal{S}_6$ and write $\text{Cr}(P) = \{Q_1, Q_2, Q_3\}$. The residue is

$$R_P^{(1)} := \text{Res}_{P=0} C_6^{(2)} = C_{5;P}^{(1)} + \sum_{Q \in \text{Cr}(P)} \frac{C_{5;P,Q}^{(0)}}{Q}, \quad C_{5;P,Q}^{(0)} := \text{Res}_{Q=0} \text{Res}_{P=0} C_6^{(2)}. \quad (41)$$

Here $C_{5;P}^{(1)}$ contains every remaining pole compatible with P . It is a general projected $C_5^{(1)}$; no locality assumption is made about it. The $C_{5;P,Q}^{(0)}$ are the lower, constant crossing coefficients and must be removed first.

The following table gives all six short-parent expansions. Every listed coefficient is attached to the displayed crossing pole $1/Q$, and the last column gives its strict zeros among Z_1, Z_2, Z_3, Z_4 .

parent P	crossing pole Q	lower coefficient $\sim C_5^{(0)}$	strict zeros
X_{13}	X_{24}	$C_{5;13,24}^{(0)} = \text{Res}_{X_{24}=0} \text{Res}_{X_{13}=0} C_6^{(2)}$	Z_2, Z_4
X_{13}	X_{25}	$C_{5;13,25}^{(0)} = \text{Res}_{X_{25}=0} \text{Res}_{X_{13}=0} C_6^{(2)}$	Z_3, Z_4
X_{13}	X_{26}	$C_{5;13,26}^{(0)} = \text{Res}_{X_{26}=0} \text{Res}_{X_{13}=0} C_6^{(2)}$	Z_1, Z_3, Z_4
X_{24}	X_{13}	$C_{5;24,13}^{(0)} = \text{Res}_{X_{13}=0} \text{Res}_{X_{24}=0} C_6^{(2)}$	Z_2, Z_4
X_{24}	X_{35}	$C_{5;24,35}^{(0)} = \text{Res}_{X_{35}=0} \text{Res}_{X_{24}=0} C_6^{(2)}$	Z_1, Z_3
X_{24}	X_{36}	$C_{5;24,36}^{(0)} = \text{Res}_{X_{36}=0} \text{Res}_{X_{24}=0} C_6^{(2)}$	Z_1, Z_4
X_{35}	X_{14}	$C_{5;35,14}^{(0)} = \text{Res}_{X_{14}=0} \text{Res}_{X_{35}=0} C_6^{(2)}$	Z_1, Z_2
X_{35}	X_{24}	$C_{5;35,24}^{(0)} = \text{Res}_{X_{24}=0} \text{Res}_{X_{35}=0} C_6^{(2)}$	Z_1, Z_3
X_{35}	X_{46}	$C_{5;35,46}^{(0)} = \text{Res}_{X_{46}=0} \text{Res}_{X_{35}=0} C_6^{(2)}$	Z_1, Z_2, Z_4
X_{46}	X_{15}	$C_{5;46,15}^{(0)} = \text{Res}_{X_{15}=0} \text{Res}_{X_{46}=0} C_6^{(2)}$	Z_1, Z_2, Z_3
X_{46}	X_{25}	$C_{5;46,25}^{(0)} = \text{Res}_{X_{25}=0} \text{Res}_{X_{46}=0} C_6^{(2)}$	Z_1, Z_2, Z_3
X_{46}	X_{35}	$C_{5;46,35}^{(0)} = \text{Res}_{X_{35}=0} \text{Res}_{X_{46}=0} C_6^{(2)}$	Z_1, Z_2, Z_4
X_{15}	X_{26}	$C_{5;15,26}^{(0)} = \text{Res}_{X_{26}=0} \text{Res}_{X_{15}=0} C_6^{(2)}$	Z_2, Z_3, Z_4
X_{15}	X_{36}	$C_{5;15,36}^{(0)} = \text{Res}_{X_{36}=0} \text{Res}_{X_{15}=0} C_6^{(2)}$	Z_1, Z_2, Z_3, Z_4
X_{15}	X_{46}	$C_{5;15,46}^{(0)} = \text{Res}_{X_{46}=0} \text{Res}_{X_{15}=0} C_6^{(2)}$	Z_1, Z_2, Z_3
X_{26}	X_{13}	$C_{5;26,13}^{(0)} = \text{Res}_{X_{13}=0} \text{Res}_{X_{26}=0} C_6^{(2)}$	Z_1, Z_3, Z_4
X_{26}	X_{14}	$C_{5;26,14}^{(0)} = \text{Res}_{X_{14}=0} \text{Res}_{X_{26}=0} C_6^{(2)}$	Z_2, Z_3, Z_4
X_{26}	X_{15}	$C_{5;26,15}^{(0)} = \text{Res}_{X_{15}=0} \text{Res}_{X_{26}=0} C_6^{(2)}$	Z_2, Z_3, Z_4

Every crossing coefficient has at least one strict zero, so $V_5(0)$ kills all of them. Reversed short-short entries are the same double residue; they are shown twice only because they occur in two different parent expansions.

Step 2: eliminate complete short-parent residues in order. After Step 1 we have

$$R_P^{(1)} = C_{5;P}^{(1)} \quad (P \in \mathcal{S}_6), \quad (42)$$

because the crossing terms in (41) are zero. This fact alone does *not* activate a polluted zero. For example, $C_{5;13,26}^{(0)} = 0$ does not make Z_1 usable for $C_{5;13}^{(1)}$. The row Z_1 is polluted by the entire X_{26} parent sector, so one must first prove

$$\text{Res}_{X_{26}=0} C_6^{(2)} = 0. \quad (43)$$

The correct elimination order is shown below. An entry in the fourth column refers only to a complete parent residue proved zero at an earlier stage.

stage	parent top piece	strict zeros	newly active only after this complete parent residue is zero	three used
0	$C_{5;15}^{(1)}$	Z_1, Z_2, Z_3, Z_4	none: good seed	Z_1, Z_2, Z_3
0	$C_{5;26}^{(1)}$	Z_2, Z_3, Z_4	none: marginal seed	Z_2, Z_3, Z_4
0	$C_{5;46}^{(1)}$	Z_1, Z_2, Z_3	none: marginal seed	Z_1, Z_2, Z_3
1	$C_{5;13}^{(1)}$	Z_3, Z_4	Z_1 after $\text{Res}_{X_{26}=0} C_6^{(2)} = 0$	Z_1, Z_3, Z_4
2	$C_{5;24}^{(1)}$	Z_1, Z_4	Z_2 after $\text{Res}_{X_{13}=0} C_6^{(2)} = 0$	Z_1, Z_2, Z_4
3	$C_{5;35}^{(1)}$	Z_1, Z_2	Z_3 after $\text{Res}_{X_{24}=0} C_6^{(2)} = 0$	Z_1, Z_2, Z_3

Here is the propagation explicitly. The seed pieces $C_{5;15}^{(1)}, C_{5;26}^{(1)}, C_{5;46}^{(1)}$ already have at least three strict zeros, so $V_5(1)$ kills them. Since their crossing coefficients were killed in Step 1, this proves the three *complete* statements

$$\text{Res}_{X_{15}=0} C_6^{(2)} = \text{Res}_{X_{26}=0} C_6^{(2)} = \text{Res}_{X_{46}=0} C_6^{(2)} = 0. \quad (44)$$

Now and only now does Z_1 become active for $C_{5;13}^{(1)}$. Together with its strict rows Z_3, Z_4 , it supplies three usable zeros, so $V_5(1)$ kills $C_{5;13}^{(1)}$ and hence the complete X_{13} parent residue. That complete statement activates Z_2 at X_{24} , which gives Z_1, Z_2, Z_4 and kills the complete X_{24} parent. Finally the complete X_{24} statement activates Z_3 at X_{35} , giving Z_1, Z_2, Z_3 and killing the complete X_{35} parent. Equivalently, the nontrivial propagation chain is

$$\text{Res}_{X_{26}=0} C_6^{(2)} = 0 \xrightarrow{Z_1} \text{Res}_{X_{13}=0} C_6^{(2)} = 0 \xrightarrow{Z_2} \text{Res}_{X_{24}=0} C_6^{(2)} = 0 \xrightarrow{Z_3} \text{Res}_{X_{35}=0} C_6^{(2)} = 0. \quad (45)$$

The already-vanishing X_{46} parent also gives the alternative activation of Z_4 at X_{35} , but it is not needed in the displayed chain.

Combining the seed and propagated parents gives

$$\text{Res}_{P=0} C_6^{(2)} = 0 \quad (P \in \mathcal{S}_6). \quad (46)$$

Therefore every term containing any short pole has been removed.

Step 3: propagate to the long parents. Only the long alphabet $\mathcal{L}_6 = \{X_{14}, X_{25}, X_{36}\}$ remains. Since each pair in $\mathcal{L}_6$ crosses, the reduced long-parent residues are

$$\text{Res}_{X_{14}=0} C_6^{(2)} = \frac{C_{5;14,25}^{(0)}}{X_{25}} + \frac{C_{5;14,36}^{(0)}}{X_{36}}, \quad (47)$$

$$\text{Res}_{X_{25}=0} C_6^{(2)} = \frac{C_{5;25,14}^{(0)}}{X_{14}} + \frac{C_{5;25,36}^{(0)}}{X_{36}}, \quad (48)$$

$$\text{Res}_{X_{36}=0} C_6^{(2)} = \frac{C_{5;36,14}^{(0)}}{X_{14}} + \frac{C_{5;36,25}^{(0)}}{X_{25}}. \quad (49)$$

There are only three distinct constants,

$$C_{5;14,25}^{(0)}, \quad C_{5;14,36}^{(0)}, \quad C_{5;25,36}^{(0)}, \quad (50)$$

and each has all four rows Z_1, Z_2, Z_3, Z_4 strict. Hence $V_5(0)$ kills them. All long-parent residues vanish as well. Since $C_6^{(2)}$ has degree -2 , it has no pole-free remainder. Therefore

$$\boxed{V_6(2) : \quad C_6^{(2)} = 0 \text{ from four consecutive zeros.}} \quad (51)$$

By the placement-insensitivity assumption, only the counts three and four matter; the consecutive blocks are convenient representatives.

2.5 What has been proved

At five points,

$$L_5, \quad V_5(0) \text{ with one zero}, \quad V_5(1) \text{ with three zeros} \quad (52)$$

hold. They imply at six points

$$\boxed{L_6, \quad V_6(0) \text{ with one zero}, \quad V_6(1) \text{ with three zeros}, \quad V_6(2) \text{ with four zeros.}} \quad (53)$$

Equivalently, at these two stages the simultaneous induction has the form

$$\boxed{\begin{array}{l} n-1 \text{ row zeros force } B_n^{(n-3)} \text{ to be local,} \\ \nu(0) = 1, \quad \nu(k) = k+2 \quad (1 \leq k < n-3) \end{array}} \quad (n=5,6). \quad (54)$$

The proof fixes no compatible top coefficient and uses no unitarity input.

3 What still needs proof

The only non-trivial step for n -point generalization is showing how many strict zeros are inherited given P and Q, Q' etc. Even more concretely we just need to show either sufficient strict zeros are inherited, and when not, sufficient active-saturated zeros can become available.